

Segment - 2

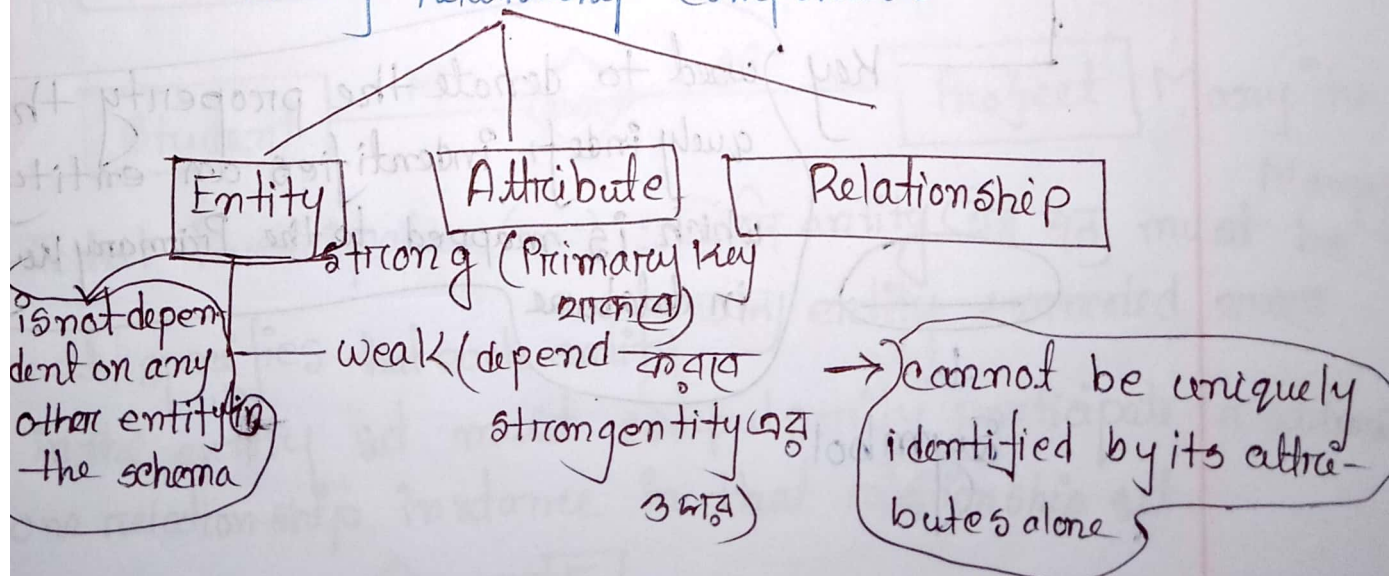
Requirements :

- Domain → Domain requirements are expectations related to a particular type of software
- Sub-Domain
- Entity Set → a real-world thing or a real-world object which is distinguishable from other objects in the real world
- Attribute → a piece of data that describes an entity
- Relationship
- Constraint : a/o

↓
primarily the way two or more data sets are linked

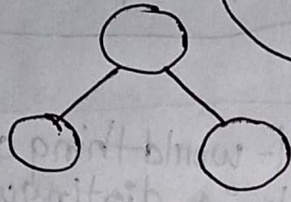
guidelines or limitations imposed on database tables, verify that data to maintain the integrity correctness and consistency of the data.

Entity Relationship Components



Attributes

composite



→ Symbol

an attribute that can be split into components is a composite

Multivalued (an attribute that can have more than one value associated with the key of the entity)



— Symbol

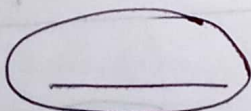
Derived a type of attribute where the value for that attribute will be derived from one or more of the other attribute of the same database



— Symbol

Key

used to denote the property that uniquely identifies an entity and which is mapped to the Primary key field in database



↓
Symbol

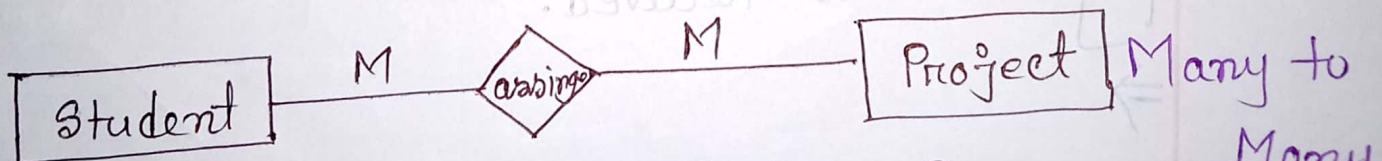
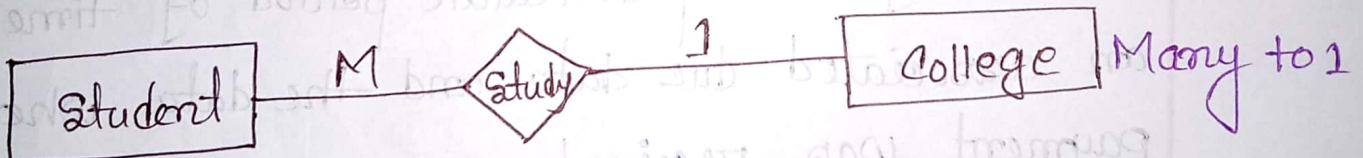
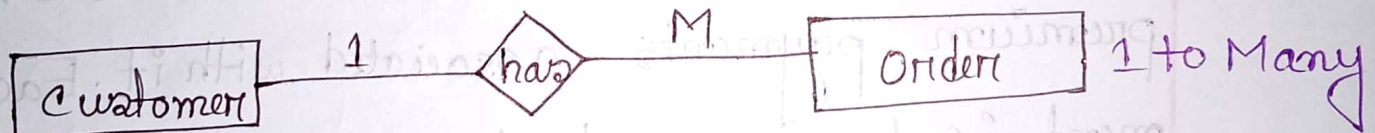
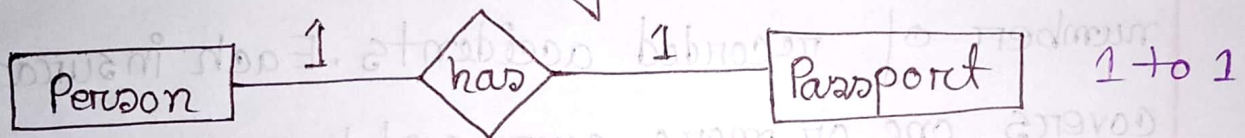
Relationship:

A relationship in DBMS is primarily the way two or more data sets are linked.

ERD - Entity relationship Diagram

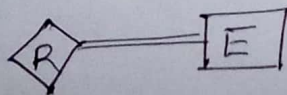
One to One relationship:

When a single instance of an entity is associated with a single instance of another entity.



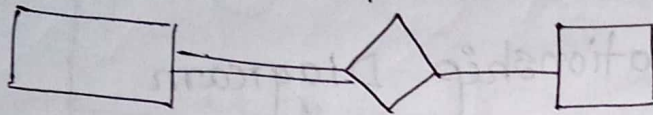
Total Participation(=) : Every entity (एक entity) must be connected (जान entity connected) to at least one entity connected.

It specifies that each entity in the entity set must compulsorily participate in at least one relationship instance in that relationship set.



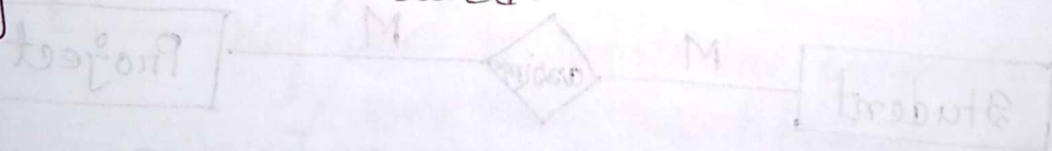
Partial Participation: (-) connected থাকতে পারে আবার
নাও থাকতে পারে।

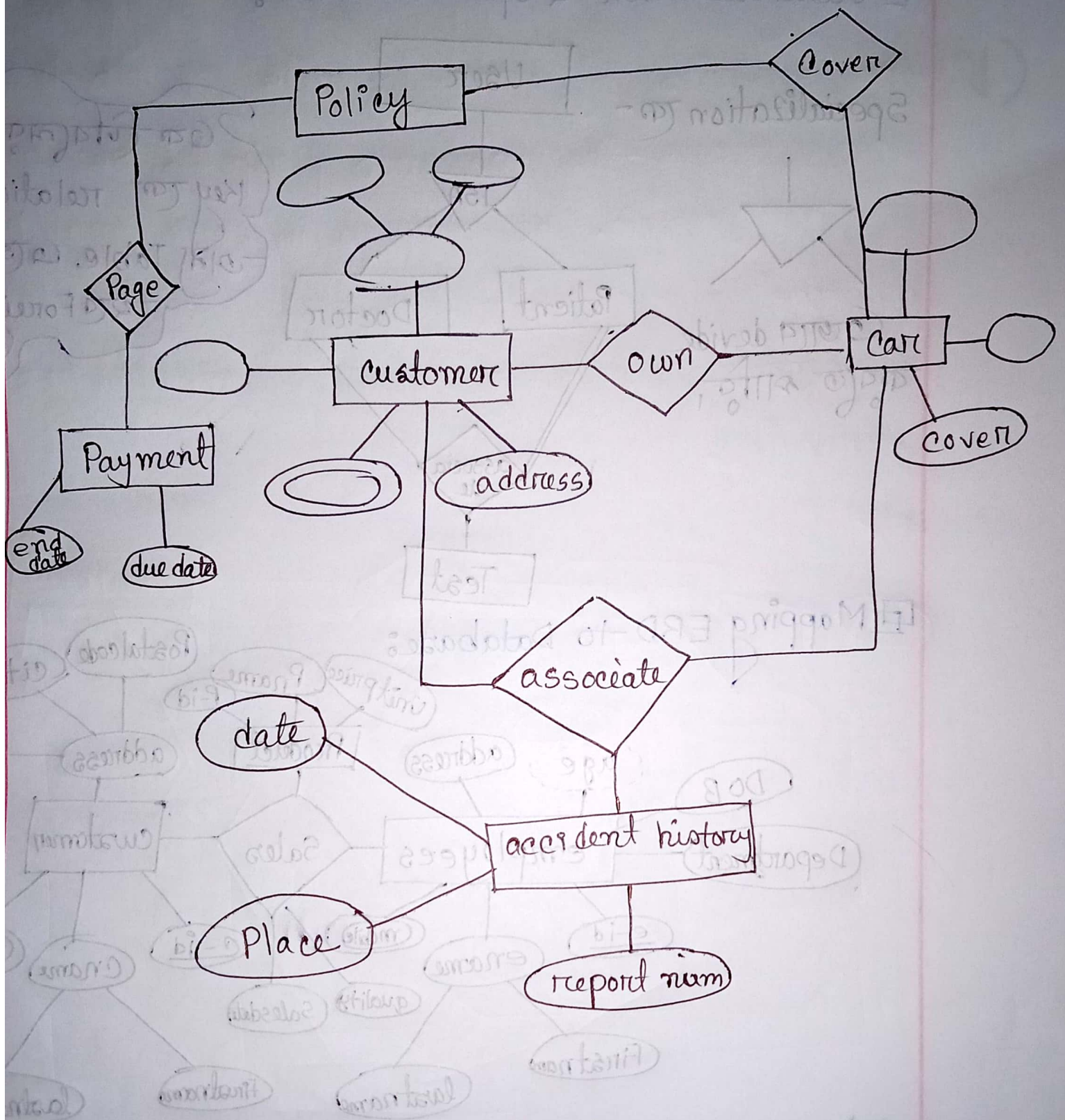
Represented using a single line between the entity
set and relationship set.



Construct an ER Diagram for a car insurance
company whose customers own one or more cars
each. Each car has associated with zero to any
number of recorded accidents. Each insurance policy
covers one or more cars and has one or more
premium payments associated with it. Each pay-
ment is for a particular period of time and has
an associated due date and the date when the
payment was received.

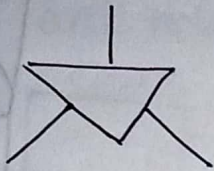
⇒



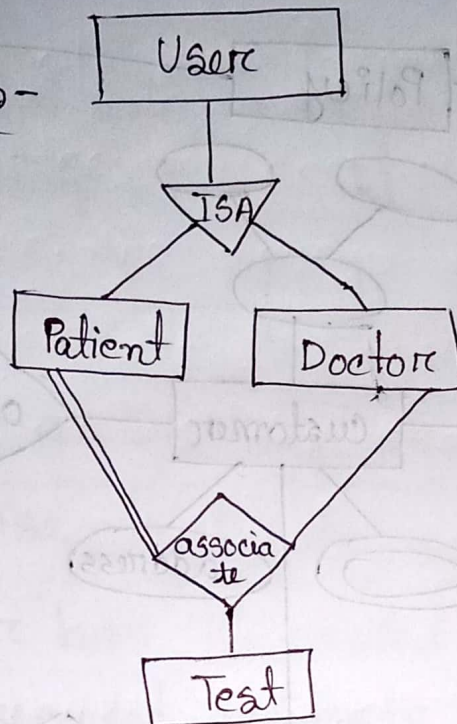


Generalization & Specialization:

Specialization ক-

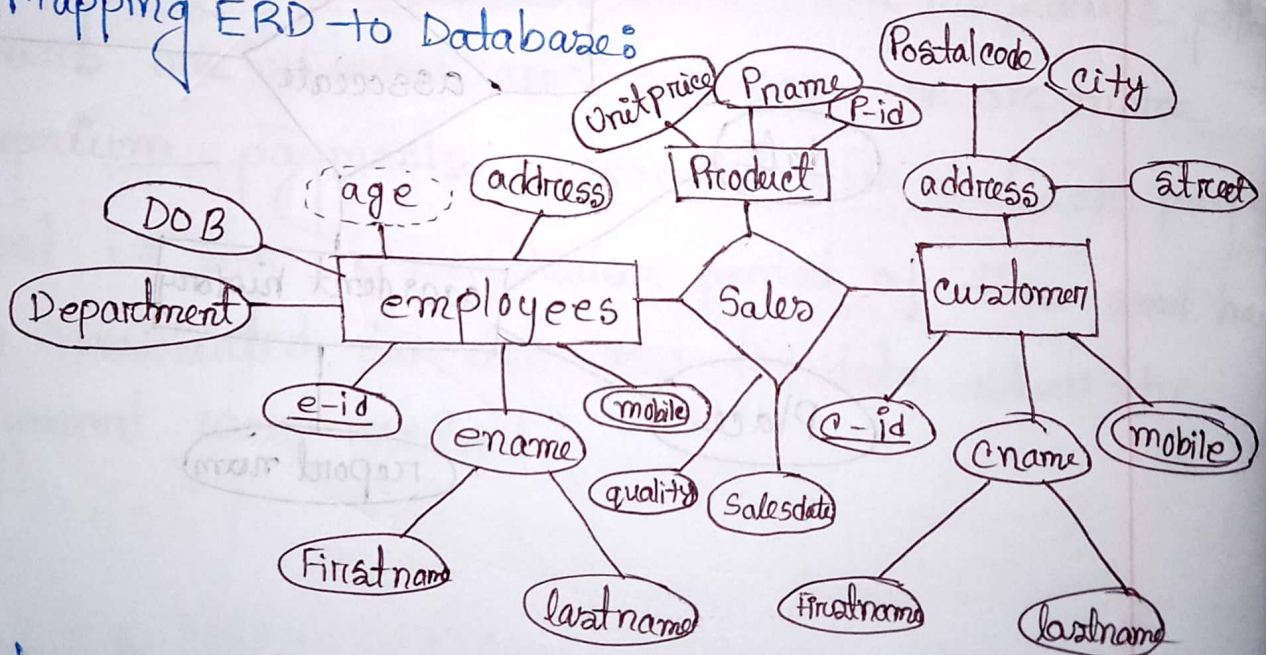


এভাবে divide
করতে পারি।



এক জীবনের Primary
Key কে relation এর
এক Table এখানে এ
হবে Foreign Key

Mapping ERD to Database:



Schema:

employees (e-id, first-name, last-name, department, mobile, DOB)

customer (c-id, first-name, last name, Postal code, city, street, mobile)

Product (P-id, p-name, unite-price)

Sales (e-id, c-id, p-id, Sales-data, quantity)

address (add-id, postal-code, city, street)

employees

e-id	First name	last name	Department	Mobile	DOB

Address

add-id	postal-code	city	street

customer

c-id	first name	last name	add-id

product

P-id	pname	unit price

Sales

e-id	c-id	P-id	Sales data	quantity

Design an E-R Diagram for keeping track of the exploits of your favorite sports team. You should store the matches played, the scores in each match, the players in each match & individual player statistics for each match. Summary statistics should be modeled as derived attributes.

